Jiahao Yao

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Education

Member of the Elite Undergraduate Program for Applied Math led by Prof. **Weinan E** — Sep. 2016-Present

B.S. Candidate in Computational Mathematics, **Peking University**, Beijing, China — Sep. 2014-Present

Academic Performance

Rank: 3/41 or 9/183 — Cumulative GPA: 3.81/4.00 (132 credits)

Core Courses

Intro to Computing	97/100	Intro to Applied Math	95.5 /100
Data Structure and Algorithm	95/100	Numerical Algebra	97/100
Numerical Analysis	97/100	Optimization	98/100

Research & Projects

Interpretation of the instability of deep learning — Independent
Supervised by Prof. Weinan E — March. 2017-Present
Methodology: Formulated neural nets as control problems, considered continuous systems.
Contributions: Related deep learning to dynamic systems, explained instability better.

Machine Learning based WRF-Chem Haze Prediction — Teamwork
CSRCC Program[1] Directed by Prof. Pingwen Zhang — March. 2016-Oct. 2017
Moderator of data group: Built a database, and carried out preprocess and postprocess of data.
Model and Algorithm: Modeled the dynamics of chemical reactions by WRF-Chem and used LSTM neural networks to make prediction.

GAN based Sharpness-aware CT Reconstruction — Independent
Supervised by Prof. Bin Dong — June. 2017-Present
Method: Unrolled traditional numerical schemes into neural networks, used *U-Nets* or *ResNets* to replace proximal opeators in primal dual proximal point methods and combined l_2 loss with GAN loss to induce sharpness of the reconstructed images.

Deep JSR Nets — Teamwork
Supervised by Prof. Bin Dong — August. 2016-Present
Method: Converted reconstruction algorithm into a multilayer ADMM net.
Model and Algorithm: Combined the idea of the joint image and spatial-Radon domain inpainting model and deep learning neural networks.

Multigrid Methods of Solving Numerical PDEs — Independent
Supervised by Prof. Jun Hu — March. 2016-Jan. 2017
Method: Faster solver to mitigate low frequency terms in anisotropic Poisson equations.
Implement: Developed faster implement of prolongation and restriction operators.

Selected Honours

2017 S.-T. Yau College Mathematics Contests, Team Gold Medal	Aug. 2017
2017 S.-T. Yau College Mathematics Contests, Silver Medal in Applied & Computational Math	Aug. 2017
2017 Mathematical Contest In Modeling, COMAP, Meritorious Winner	Apr. 2017
National Scholarship, Minister of Education (highest honor)	Oct. 2016
Pivot of Merit Student in PKU, top 1%	Oct. 2016
National Undergraduate Physics Competition, First Prize	Dec. 2015
27th Chinese Chemistry Olympiad, Silver Medal	Dec. 2013

Computation and Language Skills

Proficient: C++, Tensorflow, Pytorch
Familiar: Julia, Vim, MPI
Fluent in English: GRE: V:153, Q:170, AW:3.5

Extracurricular Activities

Service:	Chief manager of the undergraduate students' reading rooms	Sep. 2016-Present
	Chief manager of the deep learning laboratory	July. 2017-Present
Running:	Best record in 400 meter race is 56 seconds	
Teaching:	Deep Learning short course in PKU[2]	Sep.-Dec. 2016
Seminar:	Organizer of Deep Learning Seminar	Aug. 2016-Present

Miscellaneous

Science:	General Physics (I)	97/100	General Physics(II)	95/100	College Chemistry	94.5/100		
PE:	Shadowboxing	93/100	Swimming	97/100	Fencing	93/100	Table Tennis	98/100

[1]College Student Research and Career Creation Program of Beijing City
[2]course website is deeplearningpku.github.io